

Muhammad Hamza Ashraf

Fullstack Technical Lead,

A focused and diligent Software Engineer, passionate about learning new skills

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WORK EXPERIENCE

SERB, Muscat - Oman — *Senior Full-stack Engineer*

OCTOBER 2025 - PRESENT

- Architecting resilient, enterprise-grade infrastructure and modernizing legacy codebases for high-scale unmanned traffic systems.

Staff Link, Lahore — *Senior Backend Engineer - Technical Team Lead*

SEPTEMBER 2025 - PRESENT

- Technical Team Lead (Promoted from Senior Backend Engineer within 1 month)
- Leading full-stack engineering and DevOps initiatives to build scalable, secure fintech infrastructure and automate complex business workflows.

Arcompsol, Lahore — *Fractional Chief AI Officer*

NOVEMBER 2022 - PRESENT

- Advising leadership on AI strategy while shipping the technical work alongside it — not just slide decks, but the actual agentic AI pipelines, dev tooling, and team muscle to run them
- Set the AI roadmap — what to build in-house vs. buy, where LLMs add real leverage, where they'll waste budget
- Architected production agentic AI pipelines on OpenAI and Anthropic APIs — tool use, multi-turn reasoning, cost guardrails, retry logic, and prompt-injection defenses
- Stood up AI dev tooling for the engineering team — Claude Code, MCP servers, custom skills, subagent orchestration — wired into daily workflow, not bolted on as a demo
- Ran AI-readiness audits across existing services: where to inject LLMs, where to keep deterministic logic, where the security and compliance boundaries should sit
- Upskilled engineering teams through the shift from "code by hand" to "code with AI as a multiplier" — including responsible use, review discipline, and output evaluation
- Partnered with founders and executives to translate AI capability into outcomes (not hype) and to set realistic expectations with clients

Coding Crafts, Lahore — *Backend Technical Team Lead*

APRIL 2022 - AUGUST 2025

- Leading backend architecture and engineering teams to build high-performance, scalable cloud solutions.
- Specializing in database optimization, microservices design, and modernizing legacy systems for enterprise growth.
- Designing comprehensive training programs to elevate engineering standards.

Bitpix Soft, Lahore — *Backend Developer*

APRIL 2024 - AUGUST 2024

- Implemented backend features

Rolustech, Lahore — *Full Stack Developer*

SEPT 2020 - APRIL 2022

- Developed and deployed enterprise solutions
- [Customer Journey Plugin](#) for SugarCRM, enhancing CRM functionality
- [Certified Sugar Developer Specialist](#)

TECHNICAL SKILLS and TOOLS

JavaScript - TypeScript - Python - PHP - Go

NodeJS - ExpressJS - NestJS - ReactJS - NextJS - Flask - Laravel

PostgreSQL - MongoDB - AstraDB - TypeORM - Prisma - SQLAlchemy

REST - GraphQL - gRPC

Blockchain

AWS - Microsoft Azure - Docker

SOFT SKILLS

Interpersonal skills - Collaboration
Analytical skills - Problem solving
Critical thinking - Time management
Empathy & Emotional Intelligence
Leadership

CERTIFICATIONS

[Generative AI with Large Language Models](#) - AWS - [DeepLearning.ai](#)

[IBM Data Science](#)

EDUCATION

FAST NUCES, Lahore — Bachelor of Science in Computer Science
AUGUST 2016 - AUGUST 2020
CGPA 3.30

Beaconhouse School System, Lahore
— GCE Advanced Level (A-Level)
AUGUST 2014 - AUGUST 2016
Lahore — GCE Ordinary Level (O-Level)
AUGUST 2011 - AUGUST 2014

KEY PROJECTS

Interstice

March 2026 - PRESENT

- Engineered a **Canton Network ↔ Ethereum** cross-chain atomic swap engine on **DamL/HTLC** rails for regulated digital-asset settlement — hashlock-timelock contracts, reveal-based secret coordination, and dual-chain settlement verification.
- Resolved urgent production outage in **Canton JSON API v2** where a **200-element** active-contract-set (ACS) cap triggered **413** failures and cascading **SWAP_VERIFICATION_FAILED** errors — migrated ACS verification from **HTTP polling** to **Canton WebSocket streaming**
- Authored end-to-end integration test for the offramp swap flow (**Canton → Ethereum**), proving the 6-state atomic settlement lifecycle (**HASHLOCK_SHARED → ETH_LOCKED → CANTON_LOCKED → ETH_CLAIMED → CANTON_CLAIMED → COMPLETED**) and exercising the reversed secret-reveal protocol where the institution generates the hashlock secret and reveals it via the ETH claim.
- Delivered full-stack **Liquidity Provider marketplace** on **Next.js** — 3 authenticated routes for **real-time RFQ discovery** filtered by supported trading pairs, submitted-quote status tracking, and LP pair configuration — wired to the orchestrator's **POST /api/v1/rfq/{id}/quote** with bilingual (EN/ES) coverage.
- Shipped **RFQ submission form** enhancements — anonymized-trade toggle (hides institution identity during quoting phase), preferred settlement window selector, pre-submit confirmation summary (direction, asset pair, amount, slippage, fee tier), and orchestrator-config-driven minimum-trade-size validation.

Tellory

March 2026 - PRESENT

- **Hardened platform security** by implementing **SSRF prevention** across UGC and org-agent workers, adding media URL validation, prompt injection escaping for LLM inputs, and API key exposure mitigation
- Built an **AI enrichment pipeline** for products and procedures using **web search + LLM synthesis (OpenAI)**, auto-generating descriptions, categories, and aliases for 1000s of auto-created entities
- Implemented end-to-end **content moderation system** including configurable auto-moderation thresholds, banned word/phrase blocklist enforcement, and **cost guardrails** preventing runaway AI agent loops (capping at \$0.50/item vs potential \$30 runaways)
- **Eliminated data corruption risks** by converting 6+ sequential DB writes into **atomic PostgreSQL RPC transactions** across 3 workers (UGC, editorial, org-agents), and replacing brittle retry-bypass logic with proper exponential backoff
- Reduced **technical debt** by consolidating duplicated utilities (content hashing, slug generation, mention linking) into shared packages, removing ~500 lines of dead code across 5 worker services

Serb

October 2025 - PRESENT

- Engineered an **Enterprise-Grade Message Broker System** using **RabbitMQ**, implementing **Dead Letter Queues (DLQ)**, consumer timeout protection, and automatic recovery to ensure **999%** message reliability.
- **Modernized System Architecture** by migrating a monolithic codebase to a **Monorepo** with a **Modular Design** based on **SOLID** principles, significantly improving code reusability and team velocity.
- **Optimized Payment Processing** by refactoring RPC handlers from legacy switch-case logic to a **Registry-based pattern**, reducing server round-trips and improving maintainability.
- **Enhanced DevOps Reliability** by implementing **Feature Flags** for instant roll-backs, **Health Check** integrations, and **Graceful Shutdown** support for zero-downtime deployments.
- **Designed for Scale**: Configured **RabbitMQ** for multi-node deployment and reserved queue handling, allowing the system to scale horizontally across distributed environments.

LineUp

September 2025 - PRESENT

- Led **end-to-end product delivery** using **Linear**, managing the full lifecycle from requirement gathering to core Full-Stack development (FE + BE).
- **Architected a multi-party payment platform** using **Stripe Connect**, enabling automated payouts, custom invoicing, and regulatory compliance for business partners.
- **Designed a dynamic CI/CD pipeline** capable of deploying specific branches to wildcard staging environments (e.g., test-*.domain.com), utilizing organization-level secrets for secure, scalable workflows.
- **Hardened infrastructure security** by implementing **RBAC** for **MongoDB**, restricting database users from **SSH** access and enforcing the **Principle of Least Privilege**.
- **Established observability** by configuring a **Grafana, Loki, and Prometheus** stack to monitor system health and log aggregation in real-time.

HelloGov

AUGUST 2023 - PRESENT

- **Architected a Polyglot System:** Built the backend for Expediting Services using NestJS, designing a hybrid database architecture that leveraged the strengths of PostgreSQL (relational), MongoDB, and Azure CosmosDB (document store).
- **Engineering Leadership:** Led the backend team and coordinated with DevOps to integrate Azure Cloud Services, establishing a scalable infrastructure for high-availability storage and computing.
- **Database Performance Engineering:**
 - **Query Optimization:** Reduced query execution time by shifting logic to the database layer (calculations in-query), implementing strategic indexes, and utilizing bulk upserts to minimize I/O overhead.
 - **Connection Stability:** Resolved critical PostgreSQL connectivity issues by reconfiguring connection pools, implementing idle connection release strategies, and eliminating redundant data fetches.
- **Schema Design:** Designed and optimized complex database schemas to support rapid data retrieval, ensuring data consistency across multiple database engines.

Elevate

JUNE 2025 - AUGUST 2025

- **Resolved critical scaling bottlenecks** by migrating metadata storage from the Shop level to the Product Variant level using Shopify GraphQL API.
- **Overcame metadata size constraints** by redesigning the data schema, shifting from a restricted 5-field hardcoded limit to a scalable architecture.
- **Optimized query performance** to prevent connection timeouts, enabling the system to scale from limited data to supporting 50,000+ unique data points (10 experiments x 1000 products x 5 variants).
- **Eliminated legacy technical debt** by replacing hardcoded identifiers with a dynamic GraphQL-based implementation, reducing initial payload sizes and improving data integrity.

Yoga Joint - Mariana Tek

JANUARY 2023 - May 2023

- **Data Sync:** Developed a local caching system using Webhooks to mirror 3rd-party API data, reducing external dependency.
- **Performance:** Achieved a 95%+ reduction in latency (50ms local queries vs. 2000ms API calls).
- **Scalability:** Avoided API throttling and lowered metered costs by prioritizing local database hits over network requests.
- **Reliability:** Built a fallback architecture to ensure continuous app functionality during external service downtime.

CyFy

MARCH 2024 - PRESENT

- **Designed and implemented** smart contracts in TypeScript on Hyperledger Fabric (HLF)
- **Administered** the HLF network, handling channel configuration and ledger operations

Linux Foundation - Agstack

NOV 2022 - PRESENT

- **Contributed significantly** to an open-source project under the Linux Foundation
- **Created** a microservices architecture with Single Sign-On (SSO) using JWT authentication
- **Incorporated** email services for account activation and enhanced security through API key and client secret management
- **Built** Flask REST APIs and Embedded Google S2 Geometry using the s2sphere Python library
- **Authored** comprehensive documentation and Jupyter Notebook examples for the APIs

Bewta

APRIL 2024 - AUGUST 2024

- **Engineered features** for managing appointments, incorporating complex logic for employee availability and location-specific services.
- **Executed** dynamic scheduling, conflict resolution, and real-time updates for booking, rescheduling, and cancellations.

DentalWhale

AUGUST 2022 - SEP 2023

- **Developed** solutions for managing dental lab equipment
- **Designed** a microservices architecture with Auth0 integration and middleware
- **Managed** Auth0, including custom database scripts, dashboard administration, and custom flows, rules, and hooks
- **Linked** Auth0 extensions to enhance security and functionality

CrossCheck

MAY 2022 - JAN 2023

- **Programmed** a credentials management web application
- **Implemented** backend development using the specified technologies and integrated the Braintree payment gateway